MSc Technologies and Analytics in Precision Medicine

Module: Advanced Data Analytics

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1 Introduction

This report aims to analyse an orthodontic dataset originating from a study conducted at the University of North Carolina Dental School. The data set comprises repeated measures of the distance between the pituitary and pterygomaxillary fissure obtained from skull X-rays, a standard orthodontic indicator of facial bone growth. The study followed 27 children (16 males and 11 females) from age 8 to 14, with measures taken every two years at ages 8, 10, 12, and 14.

The analysis employs linear mixed-effects models to investigate this longitudinal dental growth dataset. Specifically, the report addresses how facial bone growth (measured as distance) changes with age, and whether males and females differ in their baseline measurements and growth trajectories over time.

A mixed-effects modelling approach was selected because longitudinal data contains multiple observations from the same individuals, resulting in correlated measurements within subjects. In this context, simple linear regression is inappropriate, as it assumes independence of observations and uncorrelated residuals - assumptions which are violated by repeated measures data. Applying standard linear regression to this dataset would therefore risk inflated Type 1 error rates and false confidence in p-values.

Linear mixed-effects models overcome these limitations by incorporating both fixed effects, which describe population-level trends, and random effects, which capture subject-specific deviations. This framework allows each subject to have their own regression line, thereby preserving individual variability without overfitting.

2 Methods

2.1 Exploratory Data Analysis

All code used for the exploratory analysis is provided in the attached R script (Part 1). The dataset was imported into the R environment and its structure was examined to confirm variable types and completeness. Summary statistics, including means and standard deviations, were calculated by age and sex to describe overall growth patterns and potential sex differences.

A spaghetti plot was produced to visualise individual growth trajectories across time, with separate colours used to distinguish males and females. This allowed for an initial assessment of between-subject variability in baseline measurements and growth patterns over age.

2.2 Model Building

All model fitting and comparisons were conducted using the code provided in the attached R script (Part 2). Mixed-effects models were fitted with age specified as the time variable and subject as the grouping factor.

First, a baseline mixed-effects model was fitted including age as a fixed effect and a random intercept for subject. This model was used to assess the effect of age on distance while accounting for repeated measurements within individuals.

Next, a mixed-effects model including the age-by-sex interaction was fitted with a random intercept only. This model allows subjects to differ in their baseline distance measurements while assuming a common rate of change with age across individuals.

A final mixed-effects model including the age-by-sex interaction was then fitted with both a random intercept and a random slope for age. This specification allows subjects to vary in both their baseline measurements and their rate of change over time. Predicted trajectories from the random intercept model and the random intercept and slope model were visualised using line plots to illustrate differences in assumed subject-level growth patterns.

The two random-effects structures (random intercept only versus random intercept and random slope) were compared using Akaike's Information Criterion (AIC), with model selection validated via a likelihood ratio test using ANOVA. Additionally, variance components for the random effects were calculated to quantify between-subject variability in baseline measurements and growth rates. The intraclass correlation coefficient (ICC) was calculated for the random intercept model to quantify the proportion of total variance attributable to between-subject differences.

2.3 Model Diagnostics

All diagnostic procedures were performed using the code provided in the attached R script (Part 3). Diagnostic plots of residuals versus fitted values were examined to assess linearity, homoscedasticity, and the independence of residuals from the random effects.

Normality of the residuals was evaluated using a quantile–quantile (Q–Q) plot. In addition, a Q–Q plot of the random effects for the selected model was generated to assess whether the random effects were approximately normally distributed and independent of the residual error structure.

3 Results and Interpretation

3.1 Descriptive Statistics

Table 1 displays a summary of the data by age. Each age group has an equal number of participants (n=27). The mean increased from 22 at age 8 to 26 at age 14. Variability was broadly similar across all age groups, with a slight increase in the older age groups.

Table 1: Summary Statistics (mean, SD) by Age

Age	N	Mean	SD
8	27	22	2.4
10	27	23	2.2
12	27	25	2.8
14	27	26	2.8

Table 2 summarises the data according to sex. Males showed a slightly higher mean than females, with comparable standard deviations, although sample size differed between the two groups.

Table 2: Summary Statistics (mean, SD) by Sex

Sex	N	Mean	SD
Male	64	25	2.9
Female	44	23	2.4

Figure 1 shows individual distance trajectories across age, stratified by sex. In general, distance tended to increase with age in both groups, with most individual trajectories showing an upward trend. Female participants exhibited more gradual and consistent increases, whereas male participants showed steeper increases on average along with greater inter-individual variability. Although the slopes among female participants were generally similar, some slopes between male participants showed decreases between adjacent ages points.

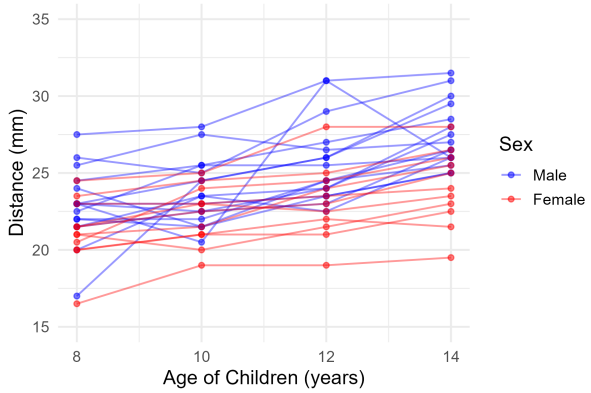

Figure 1: Longitudinal (within-subject) plot showing individual trajectories of 18 children measured across 4 time-points (8, 10, 12 and 14), stratified by sex.

3.2 Mixed-Effects Modelling

Table 3 shows that distance increases significantly with age ($\beta = 0.66$, SE $= 0.06$, $p < 0.001$) indicating an average increases of 0.66 mm per year of age, while allowing for subject-specific baseline differences.

Table 3: The Effect of Age on Distance with Random Intercepts

	Estimate	Std. Error	P.value
(Intercept)	16.76	0.80	< 0.001
Age	0.66	0.06	< 0.001

Table 4 shows that distance increases significantly with age in males ($\beta = 0.78$, SE $= 0.08$, $p < 0.001$), corresponding to an average increase of 0.78 mm per year, representing the growth rate at the population-level. At baseline, females had distances on average 1.03 higher than males, although this difference was not statistically significant ($\beta = 1.03$, SE $= 1.53$, $p > 0.05$). Additionally, there was a significant interaction between age and sex ($\beta = -0.30$, SE $= 0.12$, $p < 0.05$), indicating that the effect of age on distance was 0.30 mm smaller in females than in males. All estimates account for within-subject correlation through subject-specific random intercepts.

Table 4: The Effect of Age-Sex Interaction on Distance with Random Intercepts

	Estimate	Std. Error	P.value
(Intercept)	16.34	0.98	< 0.001
Age	0.78	0.08	< 0.001
Sex (Female)	1.03	1.53	0.50
Age $\times$ Sex (Female)	-0.30	0.12	< 0.05

A random slope for age was initially included in the model to account for individual differences in age-related change. *Table 5* presents the AIC values for the random intercept and random intercept and slope model. The random intercept model yielded a lower AIC and was therefore selected as the more parsimonious model. This decision was supported by an ANOVA comparing the two models ($\chi^2(2) = 0.83$, $p = 0.66$), indicating that allowing subject-specific age slopes did not significantly improve the fit.

Table 5: Comparison of Random Intercept and Random Intercept and Slope Model

Model	AIC
Random Intercept	440.64
Random Intercept and Random Slope	443.81

Table 6 shows the variability in baseline facial bone distance and age-related growth rates across children. Children varied considerably in baseline facial bone distance (SD $\sim$ 1.8), whereas variability in age-related growth rates was small (SD $\sim$ 0.18), supporting the selection of a random intercept model over a random intercept and slope model.

Table 6: Variance of Random Effects

Random effect	Random intercept model	Random intercept + slope model
Subject intercept	1.82	2.40
Subject age slope	–	0.18
Residual	1.39	1.31

To further illustrate the implications of the two random effects structures, *Figure 2* presents the fitted subject-level trajectories from the random intercept model (left) and the random intercept and slope model (right). In the random intercept model, subjects differ in their baseline distance measurements, but all trajectories share a common slope, resulting in parallel lines that reflect a common rate of change with age.

In contrast, the random intercept and slope model allows both baseline measurements and age-related growth rates to vary across subjects. This is reflected in the non-parallel trajectories, with individual slopes overlapping and diverging, illustrating heterogeneity in growth rates across children.

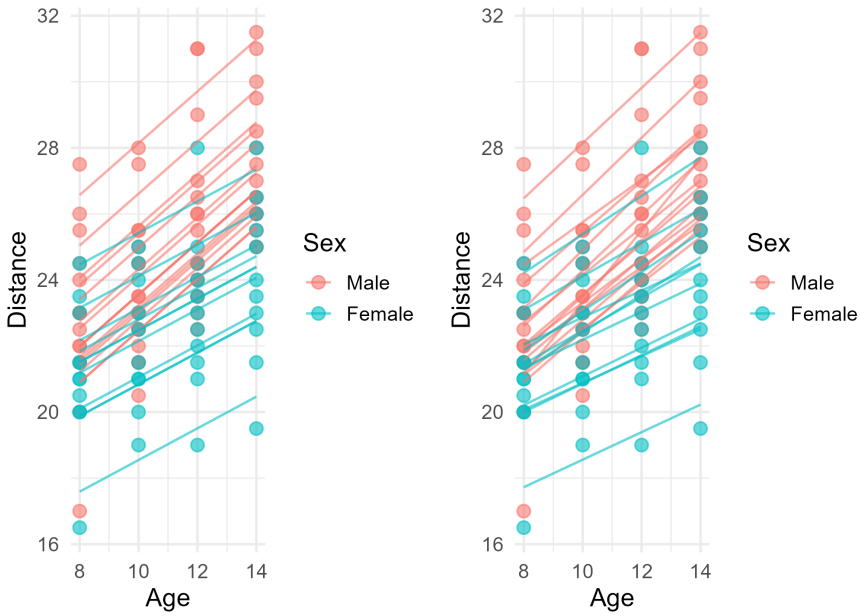

Figure 2: Random Intercepts and Random Intercepts and Slopes

The intraclass coefficient (ICC) shown in *Table 7*, quantifies the proportion of variance due to between subject differences. The adjusted ICC of the optimal model is 0.632, indicating that 63% of the variance in distance is explained by differences between subjects, thus justifying the use of a mixed-effect modelling approach.

Table 7: Intraclass Correlation Coefficient

Adjusted ICC	Unadjusted ICC
0.632	0.373

3.3 Model Diagnostics

Diagnostic plots indicated that the model assumptions were reasonably met. The residuals versus fitted values plot *(Figure 3)* showed residuals centered around zero with no strong pattern, suggesting that the linearity assumption was appropriate and that variance was approximately constant across fitted values.

The quantile-quantile (Q-Q) plot of the residuals *(Figure 4)* conveyed an agreement with the theoretical normal distribution, with only minor deviations in the lower tail, indicating approximate normality of residuals. Similarly, the Q-Q plot of the random intercepts *(Figure 5)* suggested that the random effects were approximately normally distributed, with slight deviations at the lower tail but no substantial departures from normality.

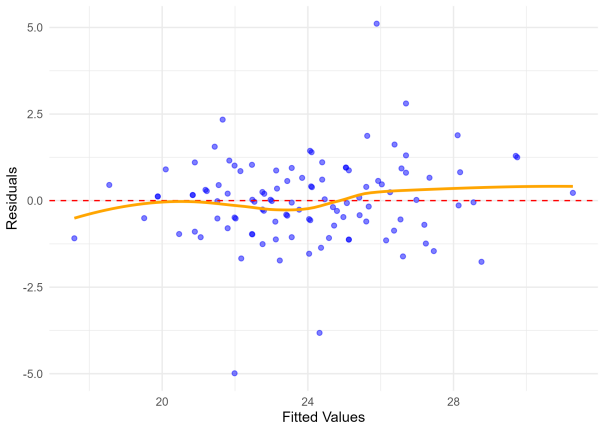

Figure 3: Plot of Residuals vs Fitted Values

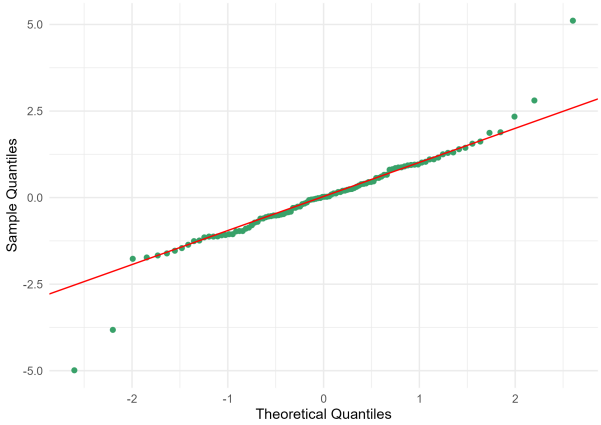

Figure 4: QQ Plot of Residuals

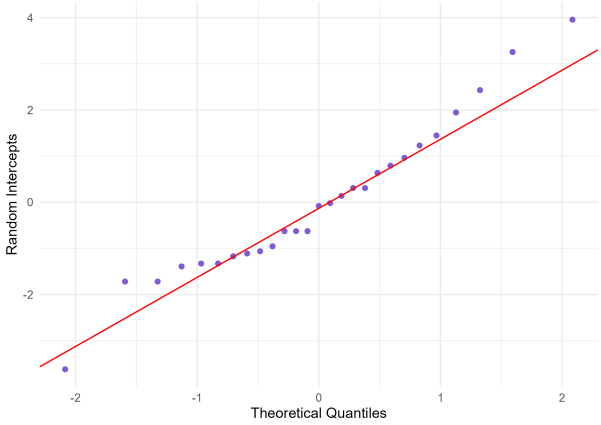

Figure 5: QQ Plot of Random Intercepts

4 Conclusion

Exploratory analysis indicated a clear increase in facial bone distance with age, consistent with a positive trajectory across childhood, with greater variability observed among males

than females.

A baseline mixed-effects model confirmed that age was a significant predictor of facial bone distance while accounting for repeated measures within individuals. Inclusion of a random intercept indicated substantial between-subject differences in baseline facial bone distance.

When sex and the age-sex interaction were included, age remained a significant predictor of distance. The main effect of sex was not statistically significant, providing no evidence of a difference in baseline facial bone distance between males and females. However, the age-sex interaction was significant and negative, indicating that the rate of increase in distance with age was smaller in females than in males.

Model comparison showed that a random intercept model was preferred over a random intercept and slope model, as allowing individual-specific growth rates did not significantly improve model fit. This suggest that while children differ in baseline facial bone distance, there is insufficient evidence that they differ substantially in their age-related growth rates beyond average sex-specific differences.

In conclusion, facial bone distance increased with age, with evidence of differing growth rates between sexes but no difference in baseline distance. It's important to note that these findings are based on a small sample size (n = 27), which may limit power to detect minor sex-specific differences.